Preliminary Implementation of Human-Swarm-Interaction Methodology with Hand-Gesture

Jinuk Heo[1], Yongseok Lee[2], and Dongjun Lee[1]

***Abstract*— We propose a novel interface to control swarm drones in 3D VR environment. The presented interface exploits dexterous hand motion to deal with large-DOF of swarm. The hand-tracker and a keyboard are used as input devices and an HMD is used for visual feedback. Various human-swarm-interaction (HSI) functionalities including virtual fixtures, way-points, and virtual sculpting are suggested and classified into three taxonomy - control, perception, and hand modulation. Furthermore, some application results such as swarm formation flight, and swarm lettering are shown using our HSI interface.**

Keywords **—** human-swarm-interaction, human-robot-interaction, user-interface, hand-tracking

I. Introduction

A swarm is a set of self-organizing agents which behaves to execute the common purpose cooperatively. In nature, for instance, a swarm of migratory birds fly in formation to conserve their energy over long distance, and a swarm of sardines swim into groups to evade predators. Moving in a swarm performs more efficiently than a single agent in terms of cost and performance time, and enables cooperative operations for transportation, fire-fighting, surveillance, and etc.[1] In addition, a swarm adapts well to the changes in the surrounding environment (adaptability), be robust to partial failure (robustness), and exhibits similar behaviors regardless of the swarm's scale (scalability)[2]. For this reason, there have been many researches on autonomous control of swarm robots since 2000s. The performance of autonomous control of swarm robots, however, still lags behind that of human in complex, dynamic environments, and furthermore, the automated control does not perfectly match to the intention of the human operator. [3], [4] So that, the user interface for human-swarm-interaction (HSI) is essentially needed, and there have been active researches on an intuitive HSI interfaces accordingly.

There have been some prior researches about HSI in 2D interface [5], [6], [7], however, these researches have definite limitations that can be applied for only 2D swarm interaction.

Most of the study about HSI in 3D environment utilizes body gesture, especially hand gesture to control high DOF system intuitively [8], [9], [10], [11], [12]. Nonetheless, some interfaces are only applicable to very small number of drones [12], [10], [9]. Other studies shows too simple operations which are far behind practical applications such as fire fighting, rescue, and surveillance [8], [11]. In this context, we suggest a novel multi-modal 3D HSI interface fully utilizing dexterous hand motion and advantages of swarm robots (adaptability, robustness, and scalability) focusing on interaction methods between human and swarm, but the additional problems such as low-level UAV control, delay in teleoperation, and subordinate PC-swarm communication architecture are somewhat excluded.

II. System Overview

The overall hardware setup is described in the Fig. 1, and comprises three main system; hand-tracking system, a

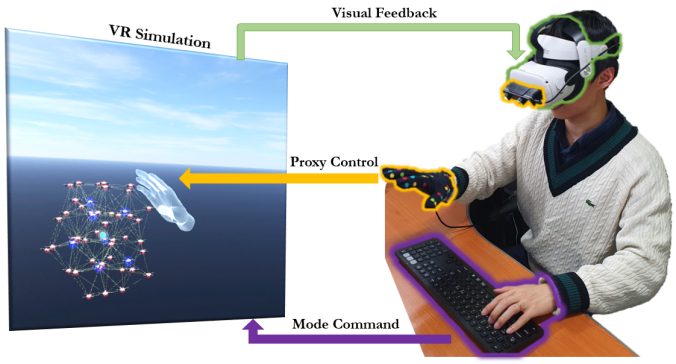


Fig. 1. System architecture of the proposed HSI interface for 3D-swarm-manipulation. The system is composed of VIST system (yellow outline), keyboard input device (purple outline), and HMD (green outline).

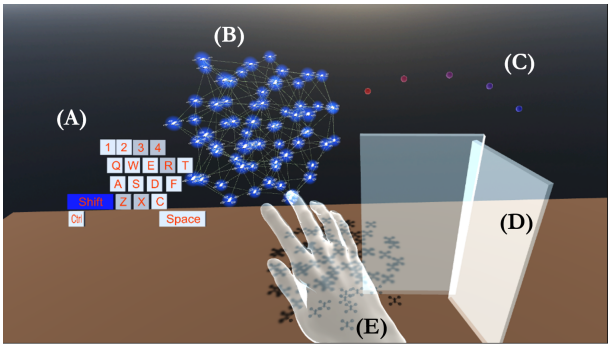


Fig. 2. The figure shows our HSI visual interface. There are five recognizable elements - (A): a virtual keyboard, (B): connectivity lines, (C): way-points (WPs), (D): virtual walls, (E): virtual hand.

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keyboard, and a head-mount-display (HMD).

• *Hand-tracking:* Hardware of hand-tracking system is composed of a sensor glove and a stereo-RGB camera (ZED mini) which are needed for VIST algorithm [13]. The VIST algorithm is based on sensor fusion of camera and IMU information in tightly-coupled manner, which enables robust hand-tracking against visual occlusion, magnetic interference, and contact which is a distinctive characteristic compared to other hand-tracking systems such as vision based tracking, IMU and magnetometer based tracking, or soft-stretch sensor based tracking.

• *Multi-modal input system:* Our multi-modal input system is combination of hand gesture and keyboard. We applied multi-modal input system considering proper operation that each dominant/nondominant hand can execute. Lots of toggling commands are even harder to be performed by hand-tracking, but non-dominant hand is also able to press predefined keys accurately and immediately.

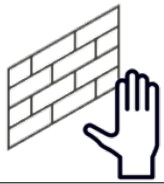
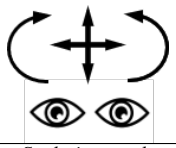
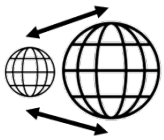
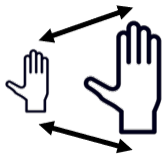
• *HMD:* We used an HMD (Oculus Quest 2) to provide the user suitable viewpoint according to its head pose estimation. In addition, HMD is essential for 3D spatial perception because it renders different images for each eye which result in the distance sensing. It is cumbersome to get distance sensing with a single 2D display because it requires persistent movement of viewpoint.

III. Functionality

Functionalities of HSI contains (control interface) virtual fixture, way-point, (perception interface) viewpoint control, map scaling, and (hand modulation) hand resizing, sculpting mode.

TABLE I

Taxonomy of HSI functionalities classified into three categories-control, perception, and hand modulation.

Taxonomy	Methodologies	
Control	*Virtual fixture*	*Way-point*
		
Perception	*Viewpoint control*	*Map scaling*
		
Hand Modulation	*Sculpting mode*	*Hand resizing*
		

A. Control Functionality

1) Virtual fixture: The interface provides the functionality of building virtual fixtures which are recognized as obstacles by each drones and apply repulsive force on them. Therefore, the virtual fixtures can play role of virtual guidance of swarm's path, and also prevent the swarm approaching particular region in the map.

2) Way-point: The presented interface has the functionality of setting serial way-points (WPs) which generate attractive force on the entire swarm. When the predefined key is pressed once, a WP is instantiated and attached on the index fingertip, then its position is fixed when the same key pressed once again. A sequence of WPs play role of swarm's path in order of creation.

B. Perception Functionality

1) Viewpoint control: The proposed HSI-UI provides the viewpoint control functionality. As normal first-person-shooter (FPS) games do, four keys of W, A, S, D takes role of moving forward, left, backward, and right. The space bar and left control key perform moving upward and downward for each. When the keys Q and E are pressed, the viewpoint turns left and right in place with respect to the global frame.

2) Map scaling: Map scaling means regulating the size of user's avatar, in other words, the user becomes a giant to oversee the entire scene or gets into the map to delicately operate the swarm. Although this functionality is classified as a perception interface but it is also useful as a control interface combined with other functionalities.

C. Hand Modulation Functionality

1) Sculpting mode: Hand-sculpting is one of the tasks which can show the dexterousness of the hand. Virtual sculpting is similar to HSI in that it requires accurate perception of space and delicate manipulation to deal with high-DOF deformable objects. There have been some prior researches on real-time virtual sculpting systems which use a haptic device [14], [15], or bare hands [16], [17] as a virtual tool. Among them, we were inspired by hand-based virtual sculpting and invented *swarm sculpting* functionality.

Swarm sculpting is, literally, shaping the formation of the swarm using the hand. Hand proxy is recognized as an obstacle during the sculpting mode, so that each drone evades it, and such behavior looks like making shape out of clay. Robust and stable hand-tracking algorithm using VIST enables dexterous hand motion to make various shapes of swarm as the Fig. 5.

2) Hand resizing: Sculpting bulky features need big hand proxy and sculpting fine features need small hand, therefore, our interface provides hand resizing functionality.

IV. Conclusion

Utilizing above functionalities, we could implement various applications of swarm control. The controller of each drone is based on the artificial potential gradient method. The Fig. 3 shows a sequential process of building virtual guidance and leading the swarm by way-points. The Fig. 4 shows



Fig. 3. Virtual fixtures work as physical guidance of the swarm.



Fig. 4. WP sequence works as a path of the swarm.

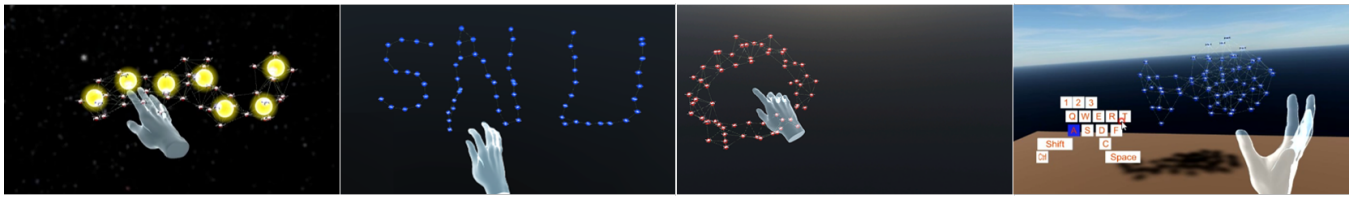


Fig. 5. The user can make various formation of drones with swarm sculpting.

a sequential process of making a path of serial way-points and the swarm following behind. The Fig. 5 shows various formation of swarm that can be built using hand sculpting mode. From the left side of the Fig. 5, the first figure shows the formation of the *Big Dipper*, the second figure shows lettering "SNU", the third figure shows a ring-shaped formation, and the last figure shows a fish-like formation.

We are going to strengthen the theoretical background of HSI-based swarm control algorithm for the future work. In addition, we aim to implement more fast, intuitive, and general HSI interface, and execute human studies.

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